

Physical world and Measurements.

Physical quantity: Anything that can be measured and quantified (expressed by a number) is known as a physical quantity.

e.g; length, mass, time, speed, velocity etc.

Beauty and intelligence are not physical quantities as they cannot be measured.

Mathematical representation of a physical quantity

$$Q = nu$$

where 'Q' is a physical quantity, 'n' is its numerical value and 'u' is its unit.

$$Q = nu$$

$$n = \frac{Q}{u}$$

$$\therefore n \propto \frac{1}{u}$$

hence, bigger the unit small is the numeric value & vice versa.

e.g $M = 1 \text{ Kg.}$
 $m = 1000 \text{ g.}$

$$L = 1 \text{ m.}$$

$$L = 100 \text{ cm.}$$

Sometimes, $Q = n$.

e.g Refractive index.
magnification.

Types of physical quantities.

- 1 Fundamental physical quantities: these are basic physical quantities. These PQs are independent of others and can be measured directly. we have 7 basic physical quantities listed below:

Mass, length, time, temperature, electric current, amount of substance, luminous intensity.

* Distance of moon from earth — 3,84,400 km

Supplementary physical quantities:

Solid angle: Angle subtended by a surface at a point (Angle subtended by a cone).

Plane angle: Angle b/w two lines.

The plane angle and the solid angle are purely mathematical physical quantities.

- 2 Derived physical quantities. These are the physical quantities which are dependent on fundamental physical quantities. eg; speed, area, velocity, acceleration, force, volume, density etc.

Units of length in astronomy

1

light year:

speed of light

$$= 3 \times 10^8 \text{ m/s}$$

$$= 3 \times 10^5 \times 10^3 \text{ m/s}$$

$$= 3 \times 10^5 \times 1000 \text{ m/s}$$

$$= 3 \times 10^5 \text{ km/s}$$

$$= 300000 \text{ km/s} = 3 \text{ lakh km/s}$$

⇒ Radius of earth = 6400 km.

It is defined as the distance travelled by light in one year.

$$\text{speed} = 3 \times 10^8 \text{ m/s}$$

$$\text{time} = 1 \text{ yr}$$

$$\text{time} = 365 \times 24 \times 60 \times 60 \text{ s}$$

$$\text{time} = 31536000 \text{ s}$$

$$\text{speed} = \frac{\text{Distance}}{\text{time}}$$

$$\text{distance} = \text{speed} \times \text{time}$$

$$1 \text{ ly} = 3 \times 10^8 \text{ m/s} \times 31536000 \text{ s}$$

$$1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$$

*

Nearest stars

Proxima centauri (4.2 ly)

Alpha centauri (4.3 ly)

2

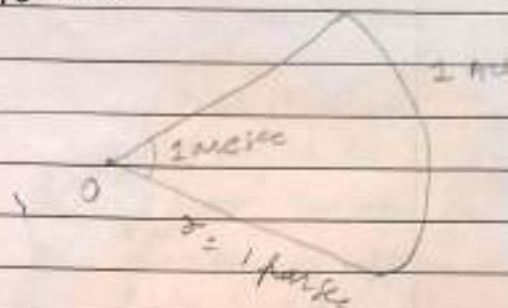
Astronomical unit (AU): It is defined as the average distance b/w the centre of sun and the centre of earth

$$1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$$

3

Par. sec (parallactic second): It is defined as the radius of that arc whose length is 1 AU and which subtends an angle of one arc second at its centre.

$$1 \text{ parsec} = 3.1 \times 10^{16} \text{ m.}$$

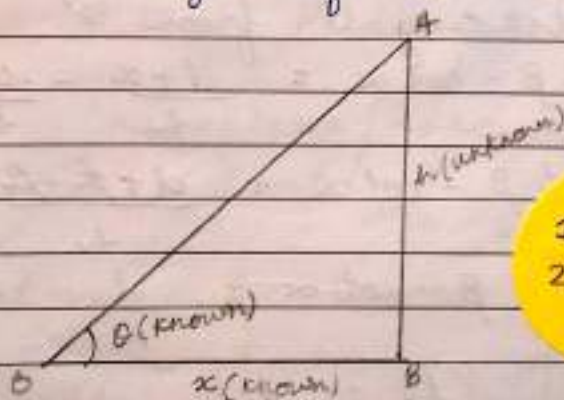


$$\therefore 1 \text{ PARSEC} > 1 \text{ LY} > 1 \text{ AU.}$$

1

Indirect methods for measuring length.
Triangulation method: This method is used to find the height of an object.

Case I: height of an accessible object.



Instruments used to measure angle of inclination

- 1) Theodolite
- 2) Clinometer / inclinometer
- 3) Sextant.

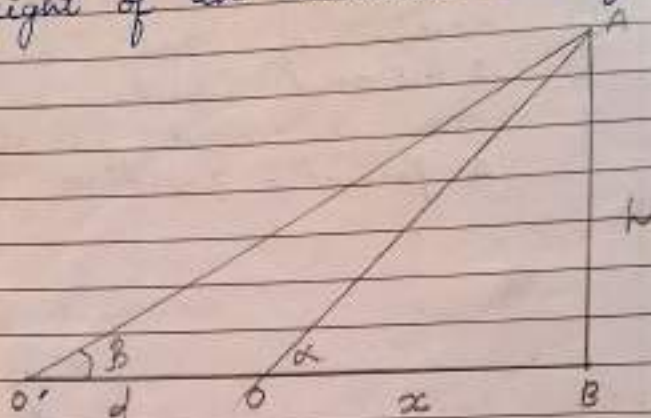
In ΔABO , we have

$$\tan \theta = \frac{AB}{BO}$$

$$\tan \theta = \frac{h}{x}$$

$$h = x \tan \theta$$

case II: height of an inaccessible object.



In $\triangle AOB$ we have,

$$\cot \alpha = \frac{x}{h} \quad \dots \text{---} (1)$$

In $\triangle AO'B$ we have,

$$\cot \beta = \frac{d+x}{h} \quad \dots \text{---} (2)$$

subtracting (1) from (2)

$$\cot \beta - \cot \alpha = \frac{d+x}{h} - \frac{x}{h}$$

$$\cot \beta - \cot \alpha = \frac{d+x-x}{h}$$

$$\cot \beta - \cot \alpha = \frac{d}{h}$$

$$h = \frac{d}{\cot \beta - \cot \alpha}$$

2 LASER method: LASER stands for Light Amplification by Stimulated Emission of Radiation

Distance travelled: $2x$

time taken = t

speed = c

$$\therefore c = \frac{\text{distance}}{\text{time}}$$

→ α -rays were discovered by Roentgen.

→ Electromagnetic waves consist of photons & travel with the speed of $3 \times 10^8 \text{ m/s}$ or $3 \times 10^{10} \text{ cm/s}$

$$c = \frac{2x}{t}$$

$$c \times t = 2x$$

$$x = \frac{c \times t}{2}$$

→ The time taken by light to reach from moon to earth is approximately 1.28 sec .
→ The distance of moon from earth is $3,84,400 \text{ km}$

3 RADAR method: RAdio Detection and Ranging. This method is used to find the dist of an aeroplane / spacecraft from an observation point.
→ same method as 2.

Electromagnetic waves

Radio waves		
Micro waves		
Infrared rays		
visible light		
ultraviolet rays		
X-rays		
Gamma rays		
	frequency increases or energy increases	wave length decreases

4 SONAR method: Sound Navigation and Ranging.

This method is used to find the distance of bodies under water.

- speed of ultrasonic waves in sea water is 1480 m/s .

5 Parallax method: This method is used to find the distance of stars from earth.

- The apparent change in the position of the object when observed from two different observation points is known as parallax.

Q1

In both the cases we use triangulation method to find the height of the building as well as the tower.
Height = distance $\times$ $\tan \theta$

Q2

Units of length in astronomy are:

i

Astronomical unit: It is the average distance b/w the sun and the earth.
 $1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$

ii

Light year: It is defined as the distance travelled by light in one year.

$$1 \text{ light year} = 9.46 \times 10^{15} \text{ m}$$

iii

Parsec: It is defined as the radius of the arc whose length is 1 AU and subtends an angle of 1 arcsecond at the centre.
 $1 \text{ Parsec} = 3.1 \times 10^{16} \text{ m}$

Descending year

$$\text{Parsec} > \text{light year} > \text{AU}$$

Q3

The two nearest stars are:

Proxima Centauri $\rightarrow$ 4.2 light years

Alpha Centauri $\rightarrow$ 4.3 light years

Time taken = 4-4.5 years.

Q4

Numerical value (n) and unit (u)

are inversely related to each other.

For eg; Length = 1 m
Length = 100 cm

- Q6 Devices used to measure angle of incline
- Ans:
i. theodolite
ii. clinometer / onclinometer.
iii. sextant.

Q6 Distance 3.84×10^8 km or 3,84,400 km
LASER method was used to find it

Q7 LASER: Light Amplification by Stimulated Emission of Radiation.

Q8 SONAR method is used to measure the depth of sea.
• sound waves are sent.
• they get reflected back.
• the time is noted and distance is calculated.

Q9 RADAR $\rightarrow$ Radio Detection And Ranging.
SONAR $\rightarrow$ Sound Navigation And Ranging.
upward

Q10 Parallax: it is the shift in position of an object when viewed from two different view points.
It is used to measure the distance of stars from earth.

Q11 It is known as Astronomical unit.
 $1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$.

Q12

Q12

Distance = 100 LY
Time taken = 100 years
b/c 1 LY is the distance
travelled by the light in 1 year.

Q13

Units of mass.
Gram, kilogram, pound, slug, amu,
quintal, metric tonne, Solar mass
and CSL (Chandra Shekhar limit.)

$$1 \text{ kg} = 1000 \text{ g}$$

$$1 \text{ pound} = 0.45 \text{ kg}$$

$$1 \text{ slug} = 14.6 \text{ kg}$$

$$1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$$

(avg mass of
p⁺ or n⁰)

$$1 \text{ quintal} = 100 \text{ kg}$$

$$1 \text{ tonne} = 1000 \text{ kg}$$

$$1 \text{ Solar mass } (M_{\odot}) = 2 \times 10^{30} \text{ kg}$$

$$1 \text{ CSL} = 1.4 \times \text{Solar mass} = 2.8 \times 10^{30} \text{ kg}.$$

Units of time.

Second, minute, hour, fortnight,
solar month, lunar month, sidereal month,
semester, lustrum, decade, indiction,
score, century and millennium.

$$1 \text{ fortnight} = 14 \text{ days}.$$

$$1 \text{ solar month} = 28 \text{ to } 31 \text{ days}.$$

$$1 \text{ lunar month} = 29.5 \text{ days}.$$

$$1 \text{ sidereal month} = 27.3 \text{ days}.$$

$$1 \text{ semester} = 6 \text{ months}.$$

$$1 \text{ lustrum} = 5 \text{ yr}.$$

$$1 \text{ decade} = 10 \text{ yr}.$$

$$1 \text{ indiction} = 15 \text{ yr}.$$

$$1 \text{ score} = 20 \text{ yr}.$$

$$1 \text{ century} = 100 \text{ yr}.$$

$$1 \text{ millennium} = 1000 \text{ yr}.$$

Moment = 90 sec

Jiffy = 10^{-2} sec

Shake = 10^{-8} sec.

Units of length.

mm, cm, m, km, mile, yard, furlong, foot, inch.

1 mm = 10^{-3} m

1 cm = 10^{-2} m

1 km = 1000 m / 10^3 m

1 mile = 1609 m.

1 inch = 2.54 cm

1 foot = 12 inch

1 yard = 3 ft.

* Length of cricket pitch = 22 yd = 66 ft.

1 furlong = 220 yd = 660 ft.

1 micrometre = 10^{-6} m

1 nanometre = 10^{-9} m

1 angstrom = 10^{-10} m

1 picometre = 10^{-12} m

1 femtometre = 10^{-15} m

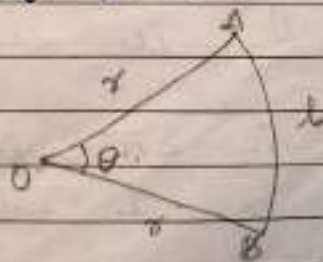
or 1 fermi

1 attometre = 10^{-18} m

Plane angle: It is defined as the angle between two lines.

Length of arc = l

radius = r .



Plane angle = $\frac{\text{length of arc}}{\text{radius of arc}}$

$$\theta = \frac{l}{r} \text{ radian} \rightarrow \text{assigned unit}$$

Angle subtended by a circle -
 $l = 2\pi r$

$$\theta = \frac{2\pi r}{r} \text{ radian}$$

$$\theta = 2\pi \text{ radian}$$

⇒ Angle subtended by a semi circle at its centre.

$$\theta = \pi \text{ radian}$$

⇒ Angle subtended by one quarter of a circle at its centre.

$$\theta = \frac{\pi}{2} \text{ radian}$$

Relation between radian and degree.

Angle subtended by a circle at its centre = 360 degree.

Angle subtended by a circle at its centre = 2π radian.

$$\therefore 2\pi \text{ radian} = 360 \text{ degree}$$

$$1 \text{ radian} = \frac{360}{2\pi} \text{ degree}$$

$$1 \text{ radian} = \frac{180}{\pi} \text{ degree or } 57.3^\circ$$

One degree (1°)

It is defined as the angle subtended by $\frac{1}{360}$ th part of a circle at its centre.

One arc minute ($1'$)

It is defined as the angle subtended by $\frac{1}{21600}$ th part of a circle at its centre.

$$1^\circ = 60 \text{ arc minute.}$$

$$1^\circ = 60'$$

One arc second ($1''$)

It is defined as the angle subtended by $\frac{1}{1296000}$ th part of a circle at its centre.

$$1 \text{ arc minute} = 60 \text{ arc Second}$$

$$1' = 60''$$

$$2\pi \text{ radian} = 360^\circ$$

$$\pi \text{ radian} = 180^\circ$$

$$\frac{\pi}{2} \text{ radian} = 90^\circ$$

$$\frac{\pi}{3} \text{ radian} = 60^\circ$$

$$\frac{\pi}{4} \text{ radian} = 45^\circ$$

$$\frac{\pi}{6} \text{ radian} = 30^\circ$$

Solid Angle.

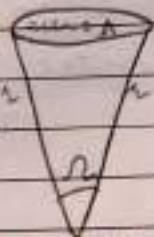
It is defined as the angle subtended by a surface at a point or

it can also be defined as the angle subtended by a cone at its vertex.

Represented by Ω (omega)

Solid angle = $\frac{\text{area of surface}}{(\text{radius})^2}$

$$\Omega = \frac{A}{r^2} \text{ steradian.}$$



Angle subtended by a sphere at its centre.

$$\frac{A}{\text{radius}^2} = \frac{4\pi r^2}{r^2} \quad (\text{area of sphere})$$

$$\therefore \Omega = \frac{A}{r^2} \text{ Sr}$$

$$\Omega = \frac{4\pi r^2}{r^2} \text{ Sr}$$

$$\therefore \text{Hemisphere} = 2\pi \text{ Sr}$$

Units of Temperature.

$$\frac{T_c}{5} = \frac{T_f - 32}{9} = \frac{T_k - 273}{5}$$

Q what is the boiling point of water in Fahrenheit scale.

$$T_C = 100^\circ\text{C}$$

$$T_F = ?$$

$$\therefore \frac{T_C}{5} = \frac{T_F - 32}{9}$$

$$\frac{100}{5} = \frac{T_F - 32}{9}$$

$$20 = \frac{T_F - 32}{9}$$

$$T_F - 32 = 9 \times 20$$

$$T_F - 32 = 180$$

$$T_F = 180 + 32 = 212^\circ\text{F}.$$

Q what is the normal ^{temp of human} body in F scale.

$$T_C = 37^\circ\text{C}.$$

$$T_F = ?$$

$$\frac{37}{5} = \frac{T_F - 32}{9}$$

$$\frac{37}{5} \times 9 = T_F - 32$$

$$66.6 = T_F - 32$$

$$66.6 + 32 = T_F$$

$$T_F = 98.6^\circ\text{F}.$$

Q convert 0°C into K and F° .

$$\rightarrow \frac{T_C}{5} = \frac{T_F - 32}{9}$$

$$\frac{0}{5} = \frac{T_F - 32}{9}$$

$$0^{\circ} = (T_F - 32)$$

$$32 = T_F$$

$$\boxed{T_F = 32^{\circ}\text{F}}$$

$$\rightarrow \frac{T_C}{5} = \frac{T_K - 273}{5}$$

$$\frac{0}{5} = \frac{T_K - 273}{5}$$

$$0 \times 5 = T_K - 273$$

$$0 + 273 = T_K$$

$$\boxed{T_K = 273\text{K}}$$

Q convert boiling pt of water into K.

$$T_C = 100^{\circ}\text{C}$$

$$T_K = ?$$

we have,

$$\frac{T_C}{5} = \frac{T_K - 273}{5}$$

$$100 = T_K - 273$$

$$100 + 273 = T_K$$

$$373 = T_K$$

$$\text{or } T_K = 373K$$

Q

At what temperature the Celsius scale and the Fahrenheit scale coincide or at what temperature the Celsius scale and the Fahrenheit ^{scale} show the same reading we have

$$\frac{T_C}{5} = \frac{T_F - 32}{9}$$

$$\text{Let } T_C = T_F = x$$

$$\therefore \frac{x}{5} = \frac{x - 32}{9}$$

$$9x = 5x - 160$$

$$9x - 5x = -160$$

$$4x = -160$$

$$x = \frac{-160}{4}$$

$$x = -40$$

system of units.

Three main physical quantities.

1 M K S system

Mass : kilogram

Length : metre

Time : second

2 C G S system

Mass : Gram

Length : centimetre

Time : second

3 F P S system

Mass : Pound

Length : Foot

Time : second

FPS system is ignored as it involves fractional values.

$$1 \text{ foot} = 0.3048 \text{ m}$$

SI system

It

1)

Mass

2)

Length

3)

Time

4)

Temperature

5)

Electricity

6)

Amount of substance

7)

Luminous intensity

8)

9)

SI system of units.

It is modified form of MKS system.

- 1) Mass $\rightarrow$ kg
- 2) length $\rightarrow$ metre
- 3) Time $\rightarrow$ second
- 4) Temperature $\rightarrow$ K
- 5) Electric current $\rightarrow$ ampere
- 6) Amount of substance $\rightarrow$ mole
- 7) Luminous intensity $\rightarrow$ candela
- 8) Plane angle $\rightarrow$ radian
- 9) Solid angle $\rightarrow$ steradian

Force.

$$F = m \cdot a$$

unit of force

$$= \text{kg m s}^{-2} = \text{newton}.$$

$$\text{Cgs} \rightarrow \text{g cm s}^{-2} = \text{dyne}.$$

$$1 \text{ newton} = 1 \text{ kg m s}^{-2}$$

$$1 \text{ newton} = 1000 \text{ g} \times 100 \text{ cm} \times \text{s}^{-2}$$

$$1 \text{ newton} = 10^3 \text{ g} \times 10^2 \text{ cm} \times \text{s}^{-2}$$

$$1 \text{ newton} = 10^5 \text{ g cm s}^{-2} = 10^5 \text{ dyne}.$$

Work.

$$W = F \cdot S$$

SI unit of work = Joule.

$$\therefore = \text{kg m}^2 \text{s}^{-2}$$

$$\begin{aligned} \text{Cgs system} &= \text{g cm}^2 \text{s}^{-2} \\ &= \text{erg} \end{aligned}$$

$$\begin{aligned}
 1 \text{ joule} &: 1 \text{ kg m}^2 \text{ s}^{-2} \\
 1 \text{ joule} &= 1000 \text{ kg} \times (100 \text{ cm})^2 \text{ s}^{-2} \\
 1 \text{ joule} &= 10^3 \text{ kg} \times 10^4 \text{ cm}^2 \text{ s}^{-2} \\
 1 \text{ joule} &= 10^7 \text{ g cm}^2 \text{ s}^{-2} \\
 1 \text{ joule} &= 10^7 \text{ erg}
 \end{aligned}$$

Dimensional analysis.

Mass M'
 Length L'
 Time T'

$$\rightarrow \text{Area} = l \times b = l \times l = L' \times L' = L'^2$$

$$\rightarrow \text{volume} = \text{length} \times \text{length} \times \text{length} = L' \times L' \times L' = L'^3$$

$$\rightarrow \text{Density} = \frac{\text{Mass}}{\text{volume}} = \frac{M'}{L'^3} = M' L'^{-3}$$

$$\rightarrow \text{speed or velocity} = \frac{\text{length}}{\text{Time}}$$

$$\text{D.F of speed or velocity} = \frac{L'}{T'} = L' T'^{-1}$$

$$\begin{aligned}
 \rightarrow \text{Acceleration} &= \frac{\text{change in velocity}}{\text{Time}} \\
 &= \frac{L' T'^{-1}}{T'} = L' T'^{-2}
 \end{aligned}$$

$$\begin{aligned}
 \rightarrow \text{Force} &= \text{mass} \times \text{acceleration} \\
 &= M' \times L' T'^{-2} = M' L' T'^{-2}
 \end{aligned}$$

$$\rightarrow \text{Pressure} = \frac{\text{Force}}{\text{area}}$$

$$\text{D.F of pressure} = \frac{M' L' T^{-2}}{L^2}$$

$$= M' L^{-1} T^{-2}$$

$$\rightarrow \text{Work} = \text{force} \times \text{displacement}$$

$$= M' L' T^{-2} \times L' = M' L^2 T^{-2}$$

$$\rightarrow \text{Kinetic energy} = \frac{1}{2} m v^2 = \frac{M' (L' T^{-1})^2}{2}$$

$$= M' L^2 T^{-2}$$

$$\rightarrow \text{Potential energy} = mgh$$

$$= M' \times L' T^{-2} \times L'$$

$$= M' L^2 T^{-2}$$

Work, KE and P.E[↑] and Torque have same dimensional formula.

$$\rightarrow \text{Torque} = \text{force} \times \text{displacement}$$

$$= M' L' T^{-2} \times L'$$

$$= M' L^2 T^{-2}$$

$$\rightarrow \text{Linear momentum} = m \cdot v$$

$$= M' \times L' T^{-1} = M' L' T^{-1}$$

$$\rightarrow \text{Angular momentum} = m \cdot v \times \text{radius}$$

$$= M' \times L' \times T^{-1} \times L'$$

$$= M' L^2 T^{-1}$$

$$\text{Power} = \frac{\text{Work}}{\text{Time}}$$

$$= \frac{M' L^2 T^{-2}}{T'}$$

$$= M' L^2 T^{-2} \times T^{-1} = M' L^2 T^{-3}$$

$$\rightarrow \text{Surface tension} = \frac{\text{force}}{\text{length}} = \frac{M' L T^{-2}}{L}$$

$$= M' T^{-2}$$

$$\rightarrow \text{Electric current} = A'$$

$$\rightarrow \text{Electric charge} = \text{current} \times \text{time}$$

$$= A' \times T'$$

$$\rightarrow \text{wave length } (\lambda) = L'$$

$$\rightarrow \text{Frequency} = \frac{1}{\text{time}} = \frac{1}{T'} = T'^{-1}$$

Dimensionless physical quantities.

1

Refractive index:

$$\text{Refractive index} = \frac{\text{speed of light in air/vacuum}}{\text{speed of light in medium}}$$
$$= \frac{c}{v}$$

$$\therefore D.F = \frac{L^1 T^{-1}}{L^1 T^{-1}} = 1 = M^0 L^0 T^0.$$

$\therefore R.I$ is dimensionless as well as unit less physical quantity.

2

Magnification:

$$= \frac{\text{height of image}}{\text{height of object}}$$

$$= \frac{L^1}{L^1} = 1 = M^0 L^0 T^0.$$

magnification is dimensionless as well as unit less phy quantity.

3

Specific gravity / relative density:

$$= \frac{\text{density of the object}}{\text{density of } H_2O \text{ at } 4^\circ C}$$

$$= \frac{M^1 L^{-3}}{M^1 L^{-3}} = 1 = M^0 L^0 T^0.$$

Specific gravity or relative density is dimensionless and unit less as well.

→ Density of H_2O is maximum at $4^\circ C$.

→ A dimensionless physical quantity may or may not have a unit.

onless physical

ex:

Index = $\frac{\text{Speed of light in a unit medium}}{\text{Speed of light in vacuum}}$

$$= \frac{c}{v}$$

$$\frac{L^1 T^{-1}}{L^1 T^{-1}} = 1 = M^0 L^0 T^0$$

is dimensionless as well as unit less physical quantity.

= height of image

4) Dielectric constant:

$$= \frac{\text{Force b/w 2 charges in free space}}{\text{Force b/w 2 charges in medium}}$$

D.F of dielectric constant.

$$= \frac{M' L' T^{-2}}{M' L' T^{-2}} = 1 = M^0 L^0 T^0$$

$\therefore$ Dielectric constant is dimensionless and unitless as well.

5) Angle:

$$\frac{\text{length of arc}}{\text{radius of arc}} = \frac{l}{r}$$

$$\therefore D.F = \frac{L'}{L'} = 1 = M^0 L^0 T^0$$

$\therefore$ Angle is dimensionless but it has a unit.

Universal Gravitational constant (G).

$$F = \frac{G m_1 m_2}{r^2}$$

$$G = \frac{F r^2}{m_1 m_2}$$

$$\therefore DF = \frac{M' L' T^{-2} \times L^2}{M' \times M'}$$

$$= M^{-1} L^3 T^{-2}$$

Plank's constant (h)

Energy of photon = $h\nu$

$$E = h\nu$$

$$h = \frac{E}{\nu}$$

$$D.F. \text{ of } h = \frac{M^1 L^2 T^{-2}}{T^{-1}}$$

$$= M^1 L^2 T^{-1}$$

and

* Angular momentum of Plank's constant have same dimensional formula.

→ Addition, subtraction, multiplication and division of Q
We can multiply and divide any two physical quantities but only those physical quantities can be added or subtracted which have same dimensional formula.

NOTE:

$$M^1 + M^1 = M^1$$

$$\text{Mass} + \text{Mass} = \text{Mass}$$

$$L^1 + L^1 = L^1$$

$$L^1 - L^1 = L^1$$

$$T^1 + T^1 = T^1$$

$$T^1 - T^1 = T^1$$

$$L^1 T^{-1} + L^1 T^{-1} = L^1 T^{-1}$$

$$L^1 T^{-1} - L^1 T^{-1} = L^1 T^{-1}$$

Q1

In the equation

$$F = \frac{a - w}{b}$$

F is force and w is work done, find the D.F. of "a" and "b".

$$\begin{aligned} \text{D.F. of } a &= \text{D.F. of } w \\ \text{D.F. of } a &= M^1 L^2 T^{-2} \end{aligned}$$

$$\text{Now, } F = \frac{a - w}{b}$$

$$\text{D.F. of } b = \frac{a - w}{F}$$

$$\text{D.F. of } b = \frac{M^1 L^2 T^{-2} - M^1 L^2 T^{-2}}{M^1 L^1 T^{-2}}$$

$$= \frac{M^1 L^2 T^{-2}}{M^1 L^1 T^{-2}} = L^1$$

Q2

In the equation $w = \frac{x - a}{b}$, w is work done, x is displacement, find the dimensional formula of a and b .

x and a will have same dimensional formula $= L^1$.

$$w = \frac{x - a}{b}$$

$$b = \frac{x - a}{w}$$

$$\text{D.F. of } b = \frac{L^1 - L^1}{M^1 L^2 T^{-2}} = \frac{L^1}{M^1 L^2 T^{-2}} = M^{-1} L^{-1} T^{+2}$$

Q3

Q4

Q3 In the equation $\left(P + \frac{a}{V^2}\right)(V - b) = RT$.

where P represents pressure and V represents volume. find the dimensional formula of a and b .

Here V and b will have same dimensional formula $= L^3$.

Now, dimensionally

$$P = \frac{a}{V^2} \text{ or } \frac{a}{V^2} = P.$$

$$\therefore a = P \times V^2.$$

$$\begin{aligned} \text{D.F of } a &= M^1 L^1 T^{-2} \times (L^3)^2 \\ &= M^1 L^1 T^{-2} \times L^6 \\ &= M^1 L^5 T^{-2}. \end{aligned}$$

Q4 In the equation $v = at^2 + bt + c$, v represents velocity and t represents time, find the dimensional formula of a , b and c .

$$\begin{aligned} \text{D.F of } c &= \text{D.F of } v \\ &= L^1 T^{-1}. \end{aligned}$$

$$\text{Dimensionally } v = at^2$$

$$\therefore \text{D.F of } a = \frac{v}{t^2}$$

$$= \frac{L^1 T^{-1}}{T^2} = L^1 T^{-1} \times T^{-2} = L^1 T^{-3}.$$

Dimensionally,

$$v = bt$$

$$\begin{aligned}\therefore \text{DF of } b &= \frac{v}{t} \\ &= \frac{L^1 T^{-1}}{T^1} = L^1 T^{-1} \times T^{-1} \\ &= L^1 T^{-2}\end{aligned}$$

Euler's number

$$e = 2.714$$

$\Rightarrow$ If there is a power to some equation - that is a physical quantity, its dimensional formula = 1.

Q.

On the equation

$$y = a e^{\frac{bx}{t^2}}$$

x is the distance and t is time.

We know,

$$\frac{bx}{t^2} = 1 \quad [\text{Dimensionally}]$$

$$\therefore b = \frac{t^2}{x}$$

$$\begin{aligned}\text{DF of } b &= \frac{T^2}{L^1} \\ &= L^{-1} T^2\end{aligned}$$

Q

The position 'x' of a particle at time 't' is given by:

$$x = \frac{V_0}{a} (1 - e^{-at^2}) \text{ where } V_0$$

is a constant, find the dimensional formula of a and V_0 .

Dimensionally

$$at^2 = 1$$

$$a = \frac{1}{t^2}$$

$$\text{DF of } a = \frac{1}{T^2} = T^{-2}$$

Dimensionally

$$x = \frac{V_0}{a}$$

$$\text{DF of } V_0 = x \times a$$

$$V_0 = L^1 \times T^{-2}$$

$$V_0 = L^1 T^{-2}$$

*

All Trigonometric functions are dimensionless.

Q.

In the equation $y = a \sin\left(\frac{bE}{t}\right)$, E represents energy and t represents time find the dimensional formula of b.

Dimensionally

$$\frac{bE}{t} = 1$$

$$\therefore b = \frac{t}{E}$$

$$\text{DF of } b = \frac{T'}{M' L^2 T^{-2}}$$

$$= M^{-1} L^{-2} T^3$$

Q In the equation $y = a e^{\frac{pb}{t}}$, p represents linear momentum and t represents time. Find the dimensional formula of b .

$$\text{Dimensionally } \frac{pb}{t} = 1$$

$$b = \frac{t}{p}$$

$$\therefore \text{DF of } b = \frac{T'}{M' L' T^{-1}}$$

$$= M^{-1} L^{-1} T^2$$

Q In the equation, $y = a \sin\left(\frac{Eb}{x}\right)$, E represents energy, and x represents distance. Find the dimensional formula of b .

$$\text{Dimensionally } \frac{Eb}{x} = 1$$

$$b = \frac{x}{E}$$

$$\text{DF of } b = \frac{L'}{M' L^2 T^{-2}}$$

$$= M^{-1} L^{-1} T^2$$

Q In the equation $y = a \sin(\omega t - kx)$, x represents distance and t represents time. Which physical quantity does $\frac{\omega}{k}$ represent.

Dimensionally,

$$\omega t = kx.$$

$$\frac{\omega}{k} = \frac{x}{t}$$

$$\therefore \text{D.F. of } \frac{\omega}{k} = \frac{L^1}{T^1} \\ = L^1 T^{-1}$$

$\therefore \frac{\omega}{k}$ represents speed or velocity.

Determination of SI and CGS units by D.F.

→ Speed = $L^1 T^{-1}$

SI unit = $m^1 \text{ sec}^{-1}$

= $m \text{ sec}^{-1}$.

CGS unit = cm sec^{-1}

→ Density = $M^1 L^{-3}$

SI unit = $\text{kg}^1 \text{ m}^{-3}$

= kg m^{-3}

CGS unit = g cm^{-3} -

Q In the equation $P = \frac{x-a}{b}$

P represents linear momentum, x represents distance, find the SI unit of b .

$$P = \frac{x-a}{b}$$

$$b = \frac{x-a}{P}$$

$$\text{DF of } b = \frac{L' - L'}{M' L' T^{-1}}$$

$$= \frac{L'}{M' L' T^{-1}} = M^{-1} T$$

$$\therefore \text{SI unit of } b = \text{kg}^{-1} \text{sec} \\ = \text{kg}^{-1} \text{sec}.$$

✓ point.

Principle of homogeneity of dimensions
It states that an equation is said to be dimensionally correct when the dimensional formula of LHS is equal to the dimensional formula of RHS.

APPLICATION 1.

checking the correctness of an equation.

e.g: $v = u + at$

DF of LHS

$$= L^1 T^{-1}$$

$$\text{DF of RHS} = L^1 T^{-1} + L^1 T^{-2} \times T^1$$

$$= L^1 T^{-1} + L^1 T^{-1}$$

$$= L^1 T^{-1}$$

As DF of LHS = DF of RHS.

$\therefore$ Acc to PHD the equation is dimensionally correct.

e.g; $S = ut + \frac{1}{2}at^2$

$$\text{DF of LHS} = L^1$$

$$\text{DF of RHS} = L^1 T^{-1} T^1 + L^1 T^{-2} T^2$$

$$= L^1 T^0 + L^1 T^0$$

$$= L^1 + L^1 = L^1$$

DF of LHS = DF of RHS.

$\therefore$ Acc to PHD, the eqn is dimensionally correct!

e.g $v^2 - u^2 = 2as$

$$\text{DF of LHS} = (L^1 T^{-1})^2 - (L^1 T^{-1})^2$$

$$= L^2 T^{-2} - L^2 T^{-2}$$

$$= L^2 T^{-2}$$

$$\text{DF of RHS} = L^1 T^{-2} \cdot L^1$$

$$= L^2 T^{-2}$$

DF of LHS = DF of RHS.

$\therefore$ Acc to PHD the eqn is dimensionally correct.

Q. check the correctness of the following equation: $T = 2\pi \sqrt{\frac{l}{g}}$.
where T is time l is length and g is acceleration due to gravity.

$$\begin{aligned} \text{DF of LHS} &= T^1 \\ \text{DF of RHS} &= \sqrt{\frac{L}{LT^{-2}}} \end{aligned}$$

$$= \sqrt{\frac{1}{T^{-2}}}$$

$$= \sqrt{T^2} = T^1$$

$$\text{DF of LHS} = \text{DF of RHS}$$

$\therefore$ Eqn is dimensionally correct.

Q. $\frac{P^2}{2m} = mgh$, where P is linear momentum, m is mass, g is a/c and h is height

$$\text{DF of LHS} = \frac{(M^1 L^1 T^{-1})^2}{M^1}$$

$$= \frac{M^2 L^2 T^{-2}}{M^1} = M^1 L^2 T^{-2}$$

$$\begin{aligned} \text{DF of RHS} &= M^1 \times L^1 T^{-2} \times L^1 \\ &= M^1 L^2 T^{-2} \end{aligned}$$

$$\text{DF of LHS} = \text{DF of RHS}$$

$\therefore$ eqn is dimensionally correct.

$$V_{\text{imp}} \cdot S_n = u + \frac{a}{2} (2n-1)$$

when S_n represents distance travelled by a particle in n^{th} second, u represents initial velocity, a represents acceleration and $2n-1$ represents time.

$$\text{DF of LHS} = L^1 T^{-1}$$

$$\begin{aligned} \text{DF of RHS} &= L^1 T^{-1} + L^1 T^{-2} \cdot T^1 \\ &= L^1 T^{-2} + L^1 T^{-1} \\ &= L^1 T^{-1} \end{aligned}$$

$$\text{DF of LHS} = \text{DF of RHS}$$

$\therefore$ Eqn is dimensionally correct.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$\therefore \text{speed} = \text{distance travelled}$$

Q

$V = \sqrt{2gh}$, where V is velocity, g is acceleration due to gravity and h is height.

$$\text{DF of LHS} = L^1 T^{-1}$$

$$\text{DF of RHS} = \sqrt{L^1 T^{-2} \cdot L^1} = \sqrt{L^2 T^{-2}}$$

$$= \sqrt{(L^1 T^{-1})^2}$$

$$= L^1 T^{-1}$$

$$\text{DF of LHS} = \text{DF of RHS}$$

$\therefore$ Eqn is dimensionally correct.



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APPLICATION 2.

Derivation of an equation.

- Q The force (F) acting on a particle moving in a circle depends on
- Mass of the particle (m)
 - Velocity of the particle (v)
 - Radius of the circle (r)
- Derive the equation for force.

$$F \propto m^a$$

$$F \propto v^b$$

$$F \propto r^c$$

$$\text{or } F = K m^a v^b r^c$$

$$F = K m^a v^b r^c \quad \text{--- (1)}$$

Dimensionally.

$$F = K m^a v^b r^c$$

$$M^1 L^1 T^{-2} = (M^1)^a (L T^{-1})^b (L^1)^c$$

$$M^1 L^1 T^{-2} = M^a L^b T^{-b} L^c$$

$$M^1 L^1 T^{-2} = M^a L^{b+c} T^{-b}$$

comparing powers.

$$\boxed{a = 1}$$

$$b = -2$$

$$\boxed{b = -2}$$

$$b + c = 1$$

$$-2 + c = 1$$

$$c = 1 - (-2)$$

$$\boxed{c = -1}$$

from equation ①

$$F = k m' v^2 r^{-1}$$

$$F = \frac{k m' v^2}{r}$$

$$F = \frac{k \cdot m v^2}{r}$$

The potential energy (U) of a particle depends on its mass (m), acceleration due to gravity (g) and its height (h) above the surface of earth.

Derive the equation for potential energy.

$$U \propto m^a$$

$$U \propto g^b$$

$$U \propto h^c$$

$$U \propto m^a g^b h^c$$

$$U = k m^a g^b h^c \text{ --- ①}$$

Dimensionally

$$m' L^2 T^{-2} = (m')^a (L' T^{-2})^b (L')^c$$

$$m' L^2 T^{-2} = m^a L^b T^{-2b} L^c$$

$$m' L^2 T^{-2} = m^a L^{b+c} T^{-2b}$$

comparing powers.

$$a = 1$$

$$- 2b = 2$$

$$b = 1$$

$$b + c = 2$$

$$1 + c = 2$$

$$c = 1$$

From eq (1)

$$U = k m^a g^b h^c$$

$$U = k m g h$$

VVI

The time period (T) of a simple pendulum depends on its length (l) and acceleration due to gravity (g).
Derive the equation for time period.

$$T \propto l^a$$

$$T \propto g^b$$

$$T \propto l^a g^b$$

$$T = k l^a g^b \quad \text{--- (1)}$$

Dimensionally

$$T' = (L')^a (g L' T'^{-2})^b$$

$$T' = L^a L^b T^{-2b}$$

$$T' = L^{a+b} T^{-2b}$$

Comparing powers.

$$-2b = 1$$

$$b = -\frac{1}{2}$$

$$a + b = 0$$

$$a + -\frac{1}{2} = 0$$

$$a = \frac{1}{2}$$

From equation ① .

$$T = k l^{\frac{1}{2}} g^{-\frac{1}{2}}$$

$$T = k \frac{l^{\frac{1}{2}}}{g^{\frac{1}{2}}}$$

$$T = k \left(\frac{l}{g} \right)^{\frac{1}{2}}$$

$$T = k \sqrt{\frac{l}{g}}$$

The velocity (V) of a particle depends on acceleration due to gravity (g) and height (h).

Derive the equation for velocity.

$$V \propto g^a$$

$$V \propto h^b$$

$$V \propto g^a h^b$$

$$V = k g^a h^b$$

//_

Dimensionally

$$L^1 T^{-1} = k g (L^1 T^{-2})^a (L^1)^b$$

$$L^1 T^{-1} = L^a T^{-2a} L^b$$

$$L^1 T^{-1} = L^{a+b} T^{-2a}$$

comparing powers.

$$-2a = -1$$

$$\boxed{a = \frac{1}{2}}$$

$$a + b = 1$$

$$\frac{1}{2} + b = 1$$

$$b = 1 - \frac{1}{2}$$

$$\boxed{b = \frac{1}{2}}$$

From equation

$$V = K g^{\frac{1}{2}} h^{\frac{1}{2}}$$

$$V = K (g h)^{\frac{1}{2}}$$

$$\boxed{V = K \sqrt{gh}}$$

//_

The acceleration due to gravity (g) depends on universal gravitational constant (G), mass of the earth (M) and radius of earth (R).

Derive the equation for acceleration due to gravity.

$$g \propto G^a$$

$$g \propto M^b$$

$$g \propto R^c$$

$$g \propto G^a M^b R^c$$

$$g = K G^a M^b R^c$$

Dimensionally

$$L^1 T^{-2} = (M^1 L^3 T^{-2})^a (M^1)^b (L^1)^c$$

$$L^1 T^{-2} = M^{-a} L^{3a} T^{-2a} M^b L^c$$

$$M^0 L^1 T^{-2} = M^{-a+b} L^{3a+c} T^{-2a}$$

comparing powers.

$$-2a = -2$$

$$\boxed{a = 1}$$

$$-a + b = 0$$

$$-1 + b = 0$$

$$\boxed{b = 1}$$

$$3a + c = 1$$

$$3 + c = 1$$

$$\boxed{c = -2}$$

From eq ①

$$g = K G^1 M^1 R^{-2}$$

$$\boxed{g = \frac{KGM}{R^2}}$$

$v \rightarrow \infty$ imp.

The wavelength (λ) of a moving particle depends on plank's constant (h), its mass (m) and its velocity (V).

Derive the equation for wavelength.

$$\lambda \propto h^a$$

$$\lambda \propto m^b$$

$$\lambda \propto v^c$$

$$\lambda \propto h^a m^b v^c$$

$$\lambda = k h^a m^b v^c$$

Dimensionally:

$$L' = (M' L^2 T^{-1})^a (M')^b (L' T^{-1})^c$$

$$M^0 L' T^0 = M^a L^{2a} T^{-a} M^b L^c T^{-c}$$

$$M^0 L' T^0 = M^{a+b} L^{2a+c} T^{-a-c}$$

Comparing powers

$$a + b = 0 \quad \text{--- } \alpha$$

$$2a + c = 1 \quad \text{--- } \beta$$

$$-a - c = 0 \quad \text{--- } \gamma$$

Adding β and γ

$$2a + c - a - c = 1 + 0$$

$$a = 1$$

Also from α

$$1 + b = 0$$

$$b = -1$$

From equation β

$$2(a) + c = 1$$

$$2 + c = 1$$

$$c = -1$$

From equation ①

$$\lambda = k h^1 m^{-1} v^{-1}$$

$$\lambda = \frac{k h}{m v}$$

$$\lambda = \frac{k \cdot h}{m v}$$

Q. 9 mp

If force (F), acceleration (A) and time (T) are taken as fundamental physical quantities. What will be the dimensional formula of length?

$$\text{DF of length} = F^a A^b T^c$$

$$L' = F^a A^b T^c \quad \text{--- ①}$$

$$M^0 L' T^0 = (M' L' T^{-2})^a (L' T^{-2})^b (T')^c$$

$$M^0 L' T^0 = M^a L^a T^{-2a} L^b T^{-2b} T^c$$

$$M^0 L' T^0 = M^a L^{a+b} T^{-2a-2b+c}$$

comparing powers

$$\boxed{a = 0}$$

$$a + b = 1$$

$$0 + b = 1$$

$$\boxed{b = 1}$$

$$-2a - 2b + c = 0$$

$$-2(0) - 2(1) + c = 0$$

$$0 - 2 + c = 0$$

$$\boxed{c = 2}$$

From eqn ①

$$\boxed{L' = F^0 A^1 T^2}$$

If force (F), velocity (V) and time (T) are taken as fundamental units what will be the dimensional formula of mass.

$$DF \text{ of Mass} = F^a V^b T^c$$

$$M' L' T' = F^a V^b T^c \quad \text{--- ①}$$

$$M' L' T' = (M' L' T^{-2})^a (L' T^{-1})^b (T')^c$$

$$M' L' T' = M'^a L'^a T^{-2a} L'^b T^{-b} T^c$$

$$M' L' T' = M'^a L'^{a+b} T^{-2a-b+c}$$

comparing powers

$$\boxed{a = 1}$$

$$a + b = 0$$

$$1 + b = 0$$

$$\boxed{b = -1}$$

$$-2a - b + c = 0$$

$$-2(1) - (-1) + c = 0$$

$$-2 + 1 + c = 0$$

$$-1 + c = 0$$

$$\boxed{c = 1}$$

From equation ①

$$[M'] = [F] [V]^{-1} [T]$$

$V \propto$

The force (F) acting on a particle is directly proportional to the square of its velocity. Find the dimensional formula of the constant.

$$F \propto V^2$$

$$F = KV^2$$

$$K = \frac{F}{V^2}$$

$$\therefore \text{D.F. of } K = \frac{[M'] [L'] [T]^{-2}}{([L'] [T]^{-1})^2} = \frac{[M'] [L'] [T]^{-2}}{[L']^2 [T]^{-2}} = \frac{[M']}{[L']} = [M' L'^{-1}]$$

Q. Find the dimensional formula of coefficient of viscosity (η) [eta] from the following equation
 $F = 6\pi\eta r v$ where F represents force
 r represents radius and v represents velocity.

$$\eta = \frac{F}{6\pi r v}$$

$$\text{D.F. of } \eta = \frac{[M'] [L'] [T]^{-2}}{[L'] \times [L'] [T]^{-1}}$$

$$[\eta] = [M' L'^{-1} T^{-1}]$$

In the equation $F = \frac{x - a}{b}$
 F represents force and b represents distance. Find the dimensional formula of $\frac{a}{b}$.

Dimensionally $x = a$
 $L' = a$
 $\therefore a = L'$

Also $b = \frac{x - a}{F}$
 $= \frac{L' - L'}{M' L' T^{-2}}$

$= \frac{L'}{M' L' T^{-2}} = M'^{-1} T^2$

$\therefore$ DF of $\frac{a}{b} = \frac{L'}{M'^{-1} T^2}$

$= L' M' T^{-2} \text{ or } M' L' T^{-2}$

In the eqn $F = 6\pi m a x$ if represents force, 'm' represents mass 'a' represents acceleration 'x' represents distance and 'b' represents length. Check the correctness of the eqn.

DF of force = $M' L' T^{-2}$ since,
 DF of RHS = $\frac{M' L' T^{-2} L'}{L^2}$

LHS $\neq$ RHS

$\therefore$ eqn is dimensionally not correct.

DF of RHS = $\frac{M' L^2 T^{-2}}{L^2}$
 $= M' T^{-2}$

3

APPLICATION III.

conversion of one system of units into another system.

In SI system

$$Q = n_1 u_1$$

In CGS system

$$Q = n_2 u_2$$

clearly

$$n_1 u_1 = n_2 u_2$$

$$n_2 = n_1 \frac{u_1}{u_2}$$

$$n_2 = n_1 \frac{M_1^a L_1^b T_1^c}{M_2^a L_2^b T_2^c}$$

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c$$

$$n_2 = n_1 \left(\frac{\text{kg}}{\text{g}} \right)^a \left(\frac{\text{m}}{\text{cm}} \right)^b \left(\frac{\text{sec}}{\text{sec}} \right)^c$$

$$n_2 = n_1 \left(\frac{\text{kg}}{\text{g}} \right)^a \left(\frac{\text{m}}{\text{cm}} \right)^b$$

$$n_2 = n_1 \left(\frac{1000 \text{ g}}{\text{g}} \right)^a \left(\frac{100 \text{ cm}}{\text{cm}} \right)^b$$

$$n_2 = n_1 (1000)^a (100)^b$$

1 newton

Q. Convert 1N of force into dyne.

we have,

$$n_1 u_1 = n_2 u_2$$

$$n_2 = n_1 \frac{u_1}{u_2}$$

$$n_2 = 1 \frac{u_1}{u_2}$$

$$n_2 = \frac{M_1' L_1' T_1'^{-2}}{M_2' L_2' T_2'^{-2}}$$

$$n_2 = \left(\frac{M_1}{M_2} \right)' \left(\frac{L_1}{L_2} \right)' \left(\frac{T_1}{T_2} \right)^{-2}$$

$$n_2 = \left(\frac{\text{kg}}{\text{g}} \right)' \left(\frac{\text{m}}{\text{cm}} \right)' \left(\frac{1 \text{ sec}}{\text{sec}} \right)^{-2}$$

$$n_2 = \left(\frac{1000 \text{ g}}{\text{g}} \right)' \left(\frac{100 \text{ cm}}{\text{cm}} \right)' (1)^{-2}$$

$$n_2 = 1000 \times 100$$

$$n_2 = 10^3 \times 10^2 = 10^5$$

$$\therefore 1 \text{ N} = 10^5 \text{ dyne}.$$

Q. Convert 1 Joule of work into erg.

we have,

$$n_1 u_1 = n_2 u_2$$

$$n_2 = n_1 \frac{u_1}{u_2}$$

$$n_2 = 1 \frac{u_1}{u_2}$$

$$n_2 = \frac{M_1 L_1^2 T_1^{-2}}{M_2 L_2^2 T_2^{-2}}$$

$$n_2 = \left(\frac{M_1}{M_2} \right)^1 \left(\frac{L_1}{L_2} \right)^2 \left(\frac{T_1}{T_2} \right)^{-2}$$

$$n_2 = \left(\frac{\text{kg}}{\text{g}} \right)^1 \left(\frac{\text{m}}{\text{cm}} \right)^2 \left(\frac{1 \text{ sec.}}{1 \text{ sec.}} \right)^{-2}$$

$$n_2 = \frac{\text{kg}}{\text{g}} \times \frac{\text{m}^2}{\text{cm}^2}$$

$$n_2 = \frac{1000 \text{ g}}{\text{g}} \times \frac{(100 \text{ cm})^2}{\text{cm}^2}$$

$$n_2 = 1000 \times \frac{10000 \text{ cm}^2}{\text{cm}^2}$$

$$n_2 = 10^3 \times 10^4$$

$$n_2 = 10^7$$

$$\therefore 1 \text{ Joule} = 10^7 \text{ erg.}$$

— / — / —

- $v = u + 3at$ [Dimensionally correct]
[Not always correct]

e.g; h, g.

e.g; ℓ and π .

e.g; force, mass, acceleration.

e.g. refractive index, magnification, specific gravity

Errors

Acceleration due to gravity = 9.8 m/s^2 .
(True value)

measured value = 9.6 m/s^2 .

$$\text{Error} = 9.8 - 9.6 = 0.2 \text{ m/s}^2.$$

→ ~~It~~ is the difference between the true value and measured value of a physical quantity is called ~~know~~ error.

Experiment.

measured values.

$$a_1, a_2, a_3, a_4, \dots, a_n.$$

mean value (a_m)

$$a_m = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}.$$

Absolute errors.

$$\Delta a_1 = a_m - a_1$$

$$\Delta a_2 = a_m - a_2$$

$$\Delta a_3 = a_m - a_3$$

⋮

$$\Delta a_n = a_m - a_n$$

mean absolute error (Δa_m).

$$\Delta a_m = \frac{|\Delta a_1| + |\Delta a_2| + |\Delta a_3| + \dots + |\Delta a_n|}{n}$$

$$\begin{array}{r} 0.1 \\ 0 \\ -0.1 \\ \hline \end{array}$$

$$\text{Final result} = (a_m \pm \Delta a_m)$$

$$\text{Relative or Fractional error} = \frac{\Delta a_m}{a_m}$$

$$\% \text{ age error} = \frac{\Delta a_m}{a_m} \times 100\%$$

In an experiment the refractive index of a medium is found to have the values 1.4, 1.5 and 1.6. Find the percentage error involved. Measured values = 1.4, 1.5 and 1.6.

Mean value

$$1. \quad a_m = \frac{1.4 + 1.5 + 1.6}{3} = 1.5$$

Absolute errors.

$$\Delta a_1 = 0.1$$

$$\Delta a_2 = 0$$

$$\Delta a_3 = -0.1$$

Mean absolute errors (Δa_m).

$$\Delta a_m = \frac{0.1 + 0 + 0.1}{3}$$

$$\Delta a_m = \frac{0.2}{3} = 0.07$$

$$\text{Final result} = 1.5 \pm 0.07$$

$$\text{Fractional error} = \frac{0.07}{1.5} = 0.05$$

relative error

$$\begin{aligned} \% \text{ age error} &= 0.05 \times 100\% \\ &= 5\% \end{aligned}$$

Q The length of a rod is $(50 \pm 0.5) \text{ cm}$.
What is the percentage error in measuring the length.

$$\text{Length} = (50 \pm 0.5) \text{ cm}.$$

$$\% \text{ age error} = \frac{0.5 \times 100\%}{50}$$

$$= \frac{0.5 \times 100}{50} \% = 1\%$$

Q. The area of a rectangle is $(10 \pm 0.1) \text{ cm}^2$.
Find the relative error.

$$\text{Area} = (10 \pm 0.1) \text{ cm}^2$$

$$\text{Relative error} = \frac{0.1}{10} = 0.01$$

Propagation of errors.

1 Error in sum:

$$(a \pm \Delta a) \quad (b \pm \Delta b)$$

Sum

$$(Z \pm \Delta Z) = (a + b) \pm (\Delta a + \Delta b)$$

$$\text{e.g.; } (10 \pm 0.1) \text{ mm} \quad (20 \pm 0.3) \text{ mm}.$$

$$\text{Total length} = (10 + 20) \pm (0.1 + 0.3) \text{ mm}$$

$$= (30 \pm 0.4) \text{ mm}.$$

Q The masses of two bodies are $(10 \pm 0.03) \text{ g}$
and $(5 \pm 0.01) \text{ g}$.
What is the total mass?

$$\text{Total mass} = (10 + 5) \pm (0.03 + 0.01) \text{ g}$$

$$= (15 \pm 0.04) \text{ g}$$

Q The areas of two rectangles are $(50 \pm 0.2) \text{ cm}^2$
and $(30 \pm 0.01) \text{ cm}^2$. Find the total area.

$$\text{Total area} = (50 + 30) \pm (0.2 + 0.01) \text{ cm}^2$$

$$= (80 \pm 0.21) \text{ cm}^2.$$

2 Error in difference.
 $(a \pm \Delta a), (b \pm \Delta b)$.

Difference

$$(Z \pm \Delta Z) = (a - b) \pm (\Delta a + \Delta b).$$

e.g. $(20 \pm 0.4) \text{ cm}, (5 \pm 0.1) \text{ cm}$.

$$\begin{aligned} \text{Difference} &= (20 - 5) \pm (0.4 + 0.1) \text{ cm} \\ &= (15 \pm 0.5) \text{ cm}. \end{aligned}$$

Q

The temperature of body 'A' is $(70 \pm 0.7)^\circ\text{C}$ and that of body 'B' is $(40 \pm 0.2)^\circ\text{C}$.

$$\begin{aligned} \text{Difference} &= (70 - 40) \pm (0.7 + 0.2)^\circ\text{C} \\ &= (30) \pm (0.9)^\circ\text{C} \\ &= (30 \pm 0.9)^\circ\text{C}. \end{aligned}$$

3

Error in product.
 $(a \pm \Delta a), (b \pm \Delta b)$

Product

$$(Z \pm \Delta Z) = (a \pm \Delta a)(b \pm \Delta b).$$

Here,

$$Z = ab.$$

$$\frac{\Delta Z}{Z} = \frac{\Delta a}{a} + \frac{\Delta b}{b}.$$

$$\Delta Z = Z \left[\frac{\Delta a}{a} + \frac{\Delta b}{b} \right]$$

Q The length of a rectangle is $(10 \pm 0.01) \text{ cm}$ and its breadth is $(6 \pm 0.02) \text{ cm}$. Find the area of the rectangle within error limits.

$$\text{Area of rect} = l \times b$$

$$(Z \pm \Delta Z) = (a \pm \Delta a)(b \pm \Delta b)$$

$$\text{Area of rect} = (A \pm \Delta A)$$

$$= (10 \pm 0.01)(6 \pm 0.02)$$

$$A = 10 \times 6 = 60$$

$$\Delta A = A \left[\frac{\Delta a}{a} + \frac{\Delta b}{b} \right]$$

$$= 60 \left[\frac{0.01}{10} + \frac{0.02}{6} \right]$$

$$= 60 [0.001 + 0.003]$$

$$= 60 \times [0.004]$$

$$= 0.24$$

$$\therefore \text{Area of rect} = (60 \pm 0.24) \text{ cm}^2$$

Q The mass of a body is $(20 \pm 0.02) \text{ kg}$ and its acceleration is $(10 \pm 0.01) \text{ m/s}^2$. Find the force of the body within error limits.

$$\text{Force} = m \times a$$

$$\text{Force} = (F \pm \Delta F) \text{ N}$$

$$\text{we know, } F = 20 \times 10 = 200 \text{ N}$$

$$\Delta F = 200 \left[\frac{0.02}{20} + \frac{0.01}{10} \right]$$

$$\Delta F = 200 [0.001 + 0.001]$$

$$\Delta F = 200 \times 0.002 = 0.4 \text{ N}$$

$$\therefore \text{Force} = (200 \pm 0.4) \text{ N}.$$

Q. The electric current through a wire is $(5 \pm 0.02) \text{ A}$ and its resistance $(2 \pm 0.01) \Omega$. Find the potential difference b/w the ends of the wire within error limits.

$$= (V \pm \Delta V)$$

$$V = 5 \times 2 = 10 \text{ volt}.$$

$$\Delta V = 10 \left[\frac{0.02}{5} + \frac{0.001}{2} \right]$$

$$= 10 \left[0.004 + 0.0005 \right]$$

$$= 10 \left[0.0045 \right]$$

$$= 0.045.$$

$$\therefore P.D = (10 \pm 0.045) \text{ volt}$$

Q 4 Error in Division.
 $(a \pm \Delta a), (b \pm \Delta b).$

$$(Z \pm \Delta Z) = \frac{(a \pm \Delta a)}{(b \pm \Delta b)}.$$

$$\text{Here } Z = \frac{a}{b}.$$

$$\frac{\Delta Z}{Z} = \frac{\Delta a}{a} + \frac{\Delta b}{b}.$$

Q. The mass of a body is $(20 \pm 0.02) \text{ kg}$ and its volume is $(4 \pm 0.002) \text{ m}^3$. Find the density of the body within error limits.

$$\rho = \frac{\text{mass}}{\text{Volume}} = (D \pm \Delta D) \text{ kg/m}^3$$

$$\text{we know, } D = \frac{20}{4} = 5 \text{ kg/m}^3$$

$$\begin{aligned} \Delta D &= 5 \left[\frac{0.02}{20} + \frac{0.002}{4} \right] \\ &= 5 \left[0.001 + 0.0005 \right] \\ &= 5 \left[0.0015 \right] \\ &= 0.0075 \text{ kg/m}^3 \end{aligned}$$

$$\therefore \text{Density} = (5 \pm 0.0075) \text{ kg/m}^3$$

Q. A potential difference of $(10 \pm 0.1) \text{ Volt}$ is applied across the ends of a wire as a result of which a current of $(2 \pm 0.02) \text{ ampere}$ passes through it. Find the resistance of the wire within error limits.

$$V = IR$$

$$R = \frac{V}{I} = R \pm \Delta R$$

$$\text{we know, } R = \frac{10}{2} = 5 \text{ ohm}$$

$$\begin{aligned} \Delta R &= 5 \left[\frac{0.1}{10} + \frac{0.02}{2} \right] \\ &= 5 \left[0.01 + 0.01 \right] \\ &= 5 \left[0.02 \right] \\ &= 0.1 \text{ ohm} \end{aligned}$$

$$\therefore \text{Resistance} = (5 \pm 0.1) \text{ ohm}$$

V^{imp}

calculation of percentage error in an equation.

In the equation

$$P = \frac{a^2 b^3 c}{d^2}$$

If the %age error in a , b , c and d are 1%, 2%, 4% and 3% respectively.

Find the % error in P .

sol.

$$\begin{aligned}\% \text{ age error in } P &= 2(1\%) + 3(2\%) + 1(4\%) + 3(2\%) \\ &= 2\% + 6\% + 4\% + 6\% \\ &= 18\%.\end{aligned}$$

Q

In the eqn $P = \frac{a^3 b^2}{c^4}$, if the %age errors in a , b and c are 2%, 3% and 1% respectively. Find the percentage error in P .

$$\begin{aligned}\% \text{ age error in } P &= 3(2\%) + 2(3\%) + 4(1\%) \\ &= 6\% + 6\% + 4\% \\ &= 16\%.\end{aligned}$$

Q

In the equation $P = \frac{a^3 \sqrt{b}}{c \sqrt{d}}$, if the percentage errors in a , b , c and d are 1%, 2%, 3% and 4% respectively. Find the percentage error in P .

$$P = \frac{a^3 b^{\frac{1}{2}}}{c d^{\frac{1}{2}}}$$

$$\begin{aligned}\therefore \% \text{ error in } P &= 3(1\%) + \frac{1}{2}(2\%) + 1(3\%) \\ &\quad + \frac{1}{2}(4\%) \\ &= 3\% + 1\% + 3\% + 2\% \\ &= 9\%.\end{aligned}$$

→ In the eqn: $P = \frac{a^3 \sqrt{b}}{c d^{\frac{1}{2}}}$, if %age error in a, b, c and d are 1%, 2%, 3% and 4% respectively. Find the %age error in P .

$$\begin{aligned} \% \text{ age error in } P &= 3(1\%) + \frac{1}{2}(2\%) + 1(3\%) \\ &+ \frac{1}{2}(4\%) = 3\% + 1\% + 3\% + 2\% \\ &= 9\% \end{aligned}$$

→ In the eqn: $P = \frac{a^2 b^3 c}{d^2}$, if the %age errors in a, b, c and d are 3%, 2%, 4% and 1% respectively. Find the %age error in P .

$$\begin{aligned} \% \text{ age error in } P &= 2(3\%) + 3(2\%) + 1(4\%) \\ &+ 2(1\%) = 6\% + 6\% + 4\% + 2\% = 18\% \end{aligned}$$

→ In the equation: $P = \frac{a b^{-2} c^3}{d^2}$, if the %age error in a, b, c and d are 2% each find the %age error in P .

$$\begin{aligned} \% \text{ age error in } P &= 1(2\%) + 2(2\%) \\ &+ 3(2\%) + 2(2\%) \\ &= 2\% + 4\% + 6\% + 4\% = 16\% \end{aligned}$$

→ In the equation $P = \frac{5a^2 b^{-1}}{c^3}$, if the %age errors in a, b and c are 2%, 3% and 1% respectively. Find the %age error in P .

$$\begin{aligned} \% \text{ age error in } P &= 2(2\%) + 1(3\%) + 3(1\%) \\ &= 4\% + 3\% + 3\% = 10\% \end{aligned}$$

→ The % age errors in measuring the mass and velocity of a body are 3% and 2% respectively. Find the % age in its K.E.

$$E = \frac{1}{2} m v^2$$

$$\% \text{ age error in } E = 1(3) + 2(2\%) = 3\% + 4\% = 7\%$$

→ The % age error in measuring the radius of a sphere is 2%. Find the % age error in its surface area and volume.

$$\text{Surface Area} = 4\pi r^2$$

$$\% \text{ age error in surface area} = 2(2\%) = 4\%$$

$$\text{Volume} = \frac{4}{3} \pi r^3 = 3(2\%) = 6\%$$

→ In the equation $P = 3a^2 b^{\frac{3}{2}} c^{-\frac{1}{2}}$, if the % age errors in a , b , c and d are 4% each, find the % age error.

$$\% \text{ age error in } P = 2(4\%) + \frac{3}{2}(4\%) + \frac{1}{2}(4\%) + 1(4\%)$$

$$= 8\% + 6\% + 2\% + 4\% = 20\%$$